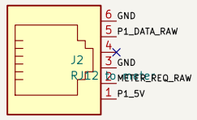
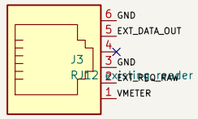
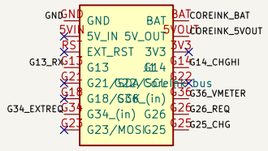
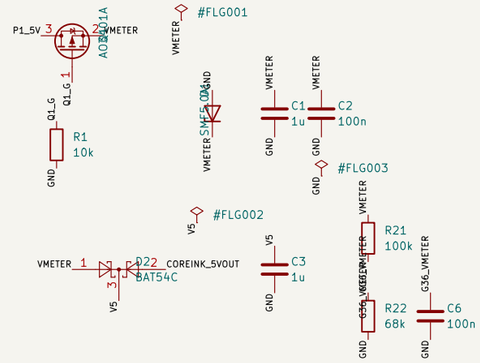
1 Connectors



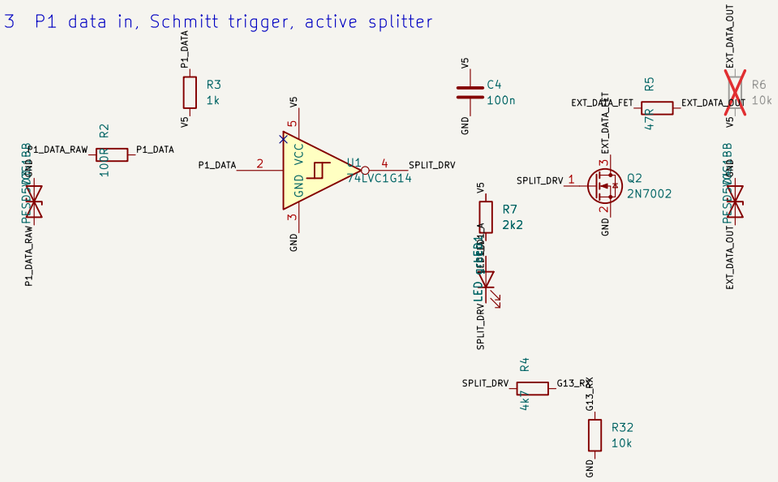




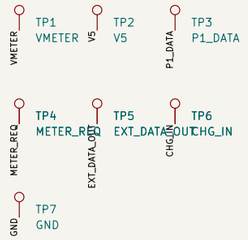
2 Meter 5 V, protection, V5



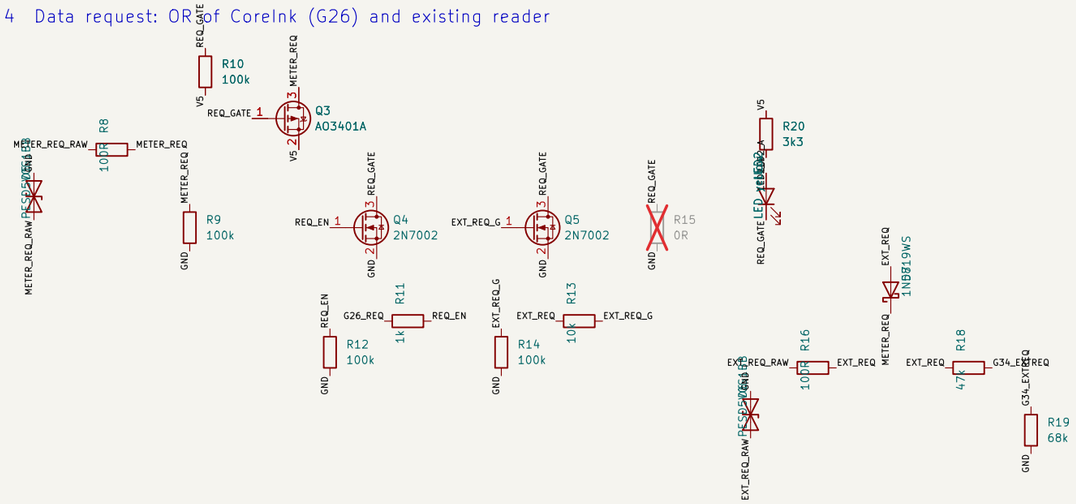
3 P1 data in, Schmitt trigger, active splitter



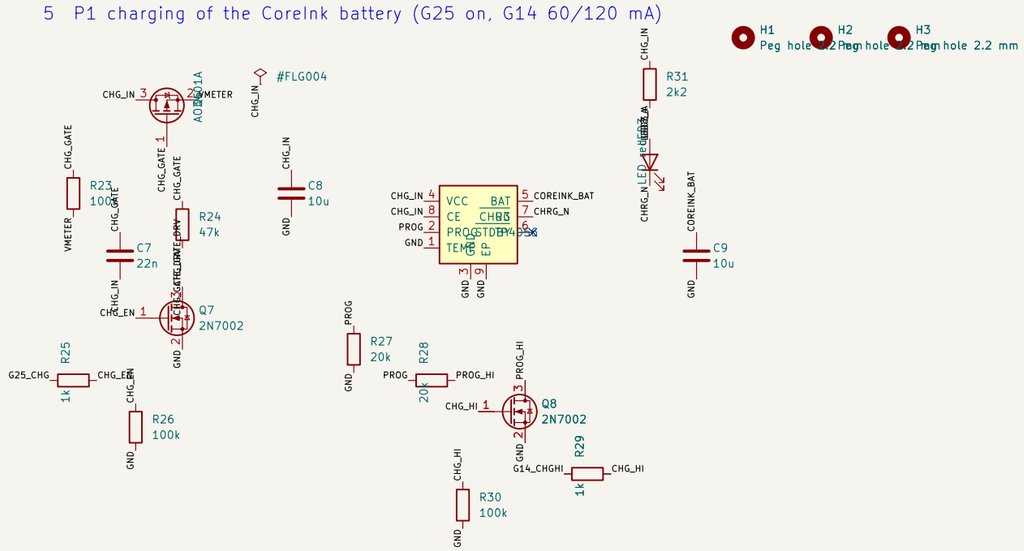
6 Test points



4 Data request: OR of CoreInk (G26) and existing reader



5 P1 charging of the CoreInk battery (G25 on, G14 60/120 mA)



7 Enclosure peg holes

H1 Peg hole 2.2 mm
H2 Peg hole 2.2 mm
H3 Peg hole 2.2 mm

NeedForHeat P1-BASE v3.0 – design rationale in docs/design-v3.md

Firmware interface: G13 = P1 RX (not inverted); G26 = data request (high = request);

G25 = P1 charging on (high); G14 = charge current 120 mA (high) / 60 mA (low);

G36 = meter 5 V via 100k/68k (x0.405); G34 = existing reader requests (47k/68k).

DNP: R6 = pull-up for readers without own pull-up; R15 = always request data.

Continues the P1-BASE by Research group Energy Transition, Windesheim
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M5Stack CoreInk extension: P1 smart meter reader with active splitter
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Title: NeedForHeat P1-BASE

Size: A3	Date: 2026-11-13	**Rev: 3.0**
KiCad E.D.A. 10.0.8		Id: 1/1